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5 / 5 submitted

Q1. Program based on Classes and Objects

Score: 3.33 **TC: 2/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollno;
    String name;
    double marks;

    void displayDetails(){
        System.out.println("Roll No: "+rollno);
        System.out.println("Name: " +name);
        System.out.println("Marks: " +marks);
    }

    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s=new Student();
        s.rollno=sc.nextInt();
        sc.nextLine();
        s.name=sc.nextLine();
        s.marks=sc.nextDouble();
        s.displayDetails();
    }
}
```

INPUT

```
16
monish
98
```

OUTPUT

```
Roll No: 16
Name: monish
Marks: 98.0
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;

    void setRollNo(int rollNo){
        this.rollNo=rollNo;
    }
    void setName(String name){
        this.name=name;
    }
    void setMarks(double marks){
        this.marks=marks;
    }
    void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s=new Student();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();

        s.setRollNo(rollNo);
        s.setName(name);
        s.setMarks(marks);

        s.displayDetails();
    }
}
```

INPUT

```
106
monish
78
```

OUTPUT

```
Roll No: 106
Name: monish
Marks: 78.0
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;

    Student(){
        rollNo=101;
        name="Rahul";
        marks=85.5;
    }

    Student(int rollNo,String name,double marks){
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;
    }

    void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s1=new Student();
        s1.displayDetails();

        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();

        Student s2=new Student(rollNo,name,marks);
        s2.displayDetails();
    }
}
```

INPUT

```
104
monish
65
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 104
Name: monish
Marks: 65.0
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Employee{
    int empId;
    String empName;
    double basicSalary;

    void getEmployeeDetails(int empId,String empName,double basicSalary){
        this.empId=empId;
        this.empName=empName;
        this.basicSalary=basicSalary;
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int empId=sc.nextInt();
        sc.nextLine();
        String empName=sc.nextLine();
        double basicSalary=sc.nextDouble();

        Salary s1=new Salary();
        s1.getEmployeeDetails(empId,empName,basicSalary);
        s1.calculateSalary();
        s1.displayDetails();
    }
}
class Salary extends Employee{
    double hra,da,netSalary;

    void calculateSalary(){
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        netSalary=basicSalary+hra+da;
    }
    void displayDetails(){
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Net Salary: "+netSalary);
    }
}
```

INPUT

```
101
Monish
50000
```

OUTPUT

```
Employee ID: 101
Employee Name:Monish
Basic Salary: 50000.0
HRA: 10000.0
DA: 5000.0
Net Salary:65000.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Employee{
    int empId;
    String empName;

    void getEmployeeDetails(int empId,String empName){
        this.empId=empId;
        this.empName=empName;
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int empId=sc.nextInt();
        sc.nextLine();
        String empName=sc.nextLine();
        double basicSalary=sc.nextDouble();

        GrossSalary g1=new GrossSalary();
        g1.getEmployeeDetails(empId,empName);
        g1.getbasicSalary(basicSalary);
        g1.calculateSalary();
        g1.displayDetails();
    }
}
class Salary extends Employee{
    double basicSalary;

    void getbasicSalary(double basicSalary){
        this.basicSalary=basicSalary;
    }
}
class GrossSalary extends Salary{
    double hra,da,grossSalary;
    void calculateSalary(){
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        grossSalary=basicSalary+hra+da;
    }

    void displayDetails(){
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Gross Salary: "+grossSalary);
    }
}
```

INPUT

```
101
monish
50000
```

OUTPUT

```
Employee ID: 101
Employee Name: monish
Basic Salary: 50000.0
HRA: 10000.0
DA: 5000.0
Gross Salary: 65000.0
```